

Lab07_DiegoLeite

Diego Leite - 182745

1. Baixe o arquivo `uwmadison.sqlite3`. Conecte-se a ele usando o pacote `RSQLite`, armazenando a conexão em uma variável `conn`.

```
library(DBI)
library(RSQLite)
```

Warning: pacote 'RSQLite' foi compilado no R versão 4.5.3

```
conn <- dbConnect(RSQLite::SQLite(), "datasets/uwmadison.sqlite3")
```

2. Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (`subjects.abbreviation='STAT'`);

```
#QUANTIDADE DE PROFESSORES
query_num_prof <- "
SELECT COUNT(DISTINCT i.id) AS total_professores
FROM instructors i
JOIN teachings t ON i.id = t.instructor_id
JOIN sections sec ON t.section_uuid = sec.uuid
JOIN subject_memberships sm ON sec.course_offering_uuid = sm.course_offering_uuid
JOIN subjects sub ON sm.subject_code = sub.code
WHERE sub.abbreviation = 'STAT';
"

num_professores <- dbGetQuery(conn, query_num_prof)
num_professores
```

```
total_professores
1                379
```

```
#LISTA DOS PROFESSORES
query_lista_prof <- "
SELECT DISTINCT i.name AS nome_professor
FROM instructors i
JOIN teachings t ON i.id = t.instructor_id
JOIN sections sec ON t.section_uuid = sec.uuid
JOIN subject_memberships sm ON sec.course_offering_uuid = sm.course_offering_uuid
JOIN subjects sub ON sm.subject_code = sub.code
WHERE sub.abbreviation = 'STAT'
ORDER BY i.name;
"

prof_stat <- dbGetQuery(conn, query_lista_prof)
head(prof_stat, 15)
```

	nome_professor
1	ABIGAIL BENZINE
2	ADIN-CRISTIAN ANDREI
3	AKICHIKA OZEKI
4	ALAN HUANG
5	ALBERTO DEL PIA
6	ALBRECHT KLEMM
7	ALEXANDER COVINGTON
8	ALEXANDER FISH
9	ALEXANDER KISELEV
10	ALYSSA DIGILIO
11	ANATOLE BECK
12	ANDREJ ZLATOS
13	ANDREW LESLIE
14	ANNE BRUCKNER
15	ANQI SHI

3. O GPA americano é definido numa escala de 0 a 4, em que $A = 4$, $AB = 3.5$, $B = 3$, $BC = 2.5$, $C = 2$, $D = 1$ e $F = 0$. Determinando a nota média de cada oferecimento pela ponderação da quantidade de alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística):

- Quem é o professor mais difícil?

```
tabela_gd <- "grade_distributions"
query_gpa_professores <- sprintf("
SELECT
    i.name AS professor,
```

```

SUM(gd.a_count * 4.0 + gd.ab_count * 3.5 + gd.b_count * 3.0 +
    gd.bc_count * 2.5 + gd.c_count * 2.0 + gd.d_count * 1.0) /
CAST(SUM(gd.a_count + gd.ab_count + gd.b_count +
    gd.bc_count + gd.c_count + gd.d_count + gd.f_count) AS FLOAT) AS media_gpa
FROM instructors i
JOIN teachings t ON i.id = t.instructor_id
JOIN sections sec ON t.section_uuid = sec.uuid
JOIN subject_memberships sm ON sec.course_offering_uuid = sm.course_offering_uuid
JOIN subjects sub ON sm.subject_code = sub.code
JOIN %s gd ON sec.course_offering_uuid = gd.course_offering_uuid
WHERE sub.abbreviation = 'STAT'
GROUP BY i.id, i.name
HAVING SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count) > 0
ORDER BY media_gpa ASC;
", tabela_gd)

gpa_profs <- dbGetQuery(conn, query_gpa_professores)

head(gpa_profs, 1)

```

```

      professor media_gpa
1 JAMES D KUELBS  2.607692

```

- Quem é o professor mais fácil?

```
tail(gpa_profs, 1)
```

```

      professor media_gpa
374 YAJUAN SI      4

```

- Qual é a disciplina mais difícil?

```

query_gpa_disciplinas <- sprintf("
SELECT
    c.number AS numero_disciplina,
    c.name AS nome_disciplina,
    SUM(gd.a_count * 4.0 + gd.ab_count * 3.5 + gd.b_count * 3.0 +
        gd.bc_count * 2.5 + gd.c_count * 2.0 + gd.d_count * 1.0) /
    CAST(SUM(gd.a_count + gd.ab_count + gd.b_count +
        gd.bc_count + gd.c_count + gd.d_count + gd.f_count) AS FLOAT) AS media_gpa
FROM courses c
JOIN course_offerings co ON c.uuid = co.course_uuid
JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
JOIN subjects sub ON sm.subject_code = sub.code

```

```

JOIN %s gd ON co.uuid = gd.course_offering_uuid
WHERE sub.abbreviation = 'STAT'
GROUP BY c.uuid, c.number, c.name
HAVING SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count) > 0
ORDER BY media_gpa ASC;
", tabela_gd)

gpa_disciplinas <- dbGetQuery(conn, query_gpa_disciplinas)

head(gpa_disciplinas, 1)

```

numero_disciplina	nome_disciplina	media_gpa
1	431 Introduction to the Theory of Probability	2.886374

- Qual é a disciplina mais fácil?

```
tail(gpa_disciplinas, 1)
```

numero_disciplina	nome_disciplina	media_gpa
70	834 Empir Proc&Semiparmtrc Infernc	4

- Desconecte do banco de dados.

```
dbDisconnect(conn)
```

```

Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()

```

```
[1] "2026-09-23 22:52:51 -03"
```